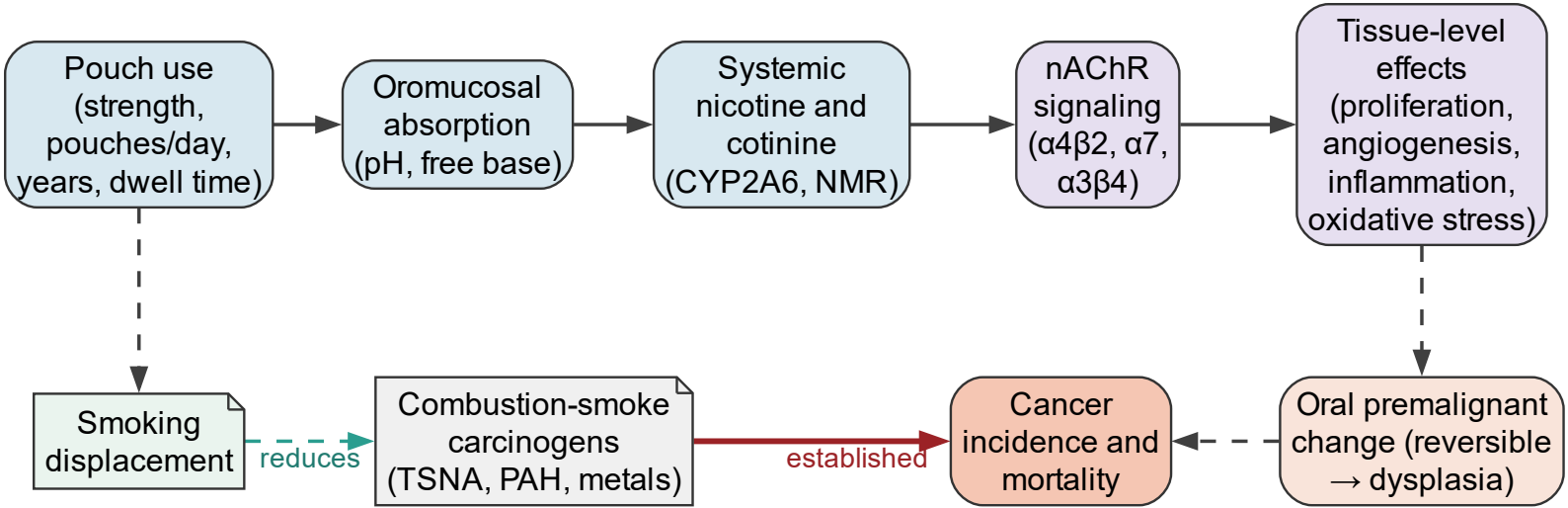
Pouch use (strength, pouches/day, years, dwell time)
Oromucosal absorption (pH, free base)
Systemic nicotine and cotinine (CYP2A6, NMR)
nAChR signaling (α4β2, α7, α3β4)
Tissue-level effects (proliferation, angiogenesis, inflammation, oxidative stress)
Smoking displacement
reduces
Combustion-smoke carcinogens (TSNA, PAH, metals)
established
Cancer incidence and mortality
Oral premalignant change (reversible → dysplasia)